

Chapter 3

Random Variables and Distributions

3.1 Random Variables and Discrete Distributions

Solutions to Exercises

1. Each of the 11 integers from 10 to 20 has the same probability of being the value of X . Six of the 11 integers are even, so the probability that X is even is $6/11$.
2. The sum of the values of $f(x)$ must be equal to 1. Since $\sum_{x=1}^5 f(x) = 15c$, we must have $c = 1/15$.
3. By looking over the 36 possible outcomes enumerated in Example 1.6.5, we find that $X = 0$ for 6 outcomes, $X = 1$ for 10 outcomes, $X = 2$ for 8 outcomes, $X = 3$ for 6 outcomes, $X = 4$ for 4 outcomes, and $X = 5$ for 2 outcomes. Hence, the p.f. $f(x)$ is as follows:

x	0	1	2	3	4	5
$f(x)$	$3/18$	$5/18$	$4/18$	$3/18$	$2/18$	$1/18$

4. For $x = 0, 1, \dots, 10$, the probability of obtaining exactly x heads is $\binom{10}{x} \left(\frac{1}{2}\right)^{10}$.
5. For $x = 2, 3, 4, 5$, the probability of obtaining exactly x red balls is $\binom{7}{x} \binom{3}{5-x} / \binom{10}{5}$.
6. The desired probability is the sum of the entries for $k = 0, 1, 2, 3, 4$, and 5 in that part of the table of binomial probabilities given in the back of the book corresponding to $n = 15$ and $p = 0.5$. The sum is 0.1509.
7. Suppose that a machine produces a defective item with probability 0.7 and produces a nondefective item with probability 0.3. If X denotes the number of defective items that are obtained when 8 items are inspected, then the random variable X will have the binomial distribution with parameters $n = 8$ and $p = 0.7$. By the same reasoning, however, if Y denotes the number of nondefective items that are obtained, then Y will have the binomial distribution with parameters $n = 8$ and $p = 0.3$. Furthermore, $Y = 8 - X$. Therefore, $X \geq 5$ if and only if $Y \leq 3$ and it follows that $\Pr(X \geq 5) = \Pr(Y \leq 3)$. Probabilities for the binomial distribution with $n = 8$ and $p = 0.3$ are given in the table in the back of the book. The value of $\Pr(Y \leq 3)$ will be the sum of the entries for $k = 0, 1, 2$, and 3.

8. The number of red balls obtained will have the binomial distribution with parameters $n = 20$ and $p = 0.1$. The required probability can be found from the table of binomial probabilities in the back of the book. Add up the numbers in the $n = 20$ and $p = 0.1$ section from $k = 4$ to $k = 20$. Or add up the numbers from $k = 0$ to $k = 3$ and subtract the sum from 1. The answer is 0.1330.
9. We need $\sum_{x=0}^{\infty} f(x) = 1$, which means that $c = 1/\sum_{x=0}^{\infty} 2^{-x}$. The last sum is known from Calculus to equal $1/(1 - 1/2) = 2$, so $c = 1/2$.
10. (a) The p.f. of X is $f(x) = c(x+1)(8-x)$ for $x = 0, \dots, 7$ where c is chosen so that $\sum_{x=0}^7 f(x) = 1$. So, c is one over $\sum_{x=0}^7 (x+1)(8-x)$, which sum equals 120, so $c = 1/120$. That is $f(x) = (x+1)(8-x)/120$ for $x = 0, \dots, 7$.
- (b) $\Pr(X \geq 5) = [(5+1)(8-5) + (6+1)(8-6) + (7+1)(8-7)]/120 = 1/3$.
11. In order for the specified function to be a p.f., it must be the case that $\sum_{x=1}^{\infty} \frac{c}{x} = 1$ or equivalently $\sum_{x=1}^{\infty} \frac{1}{x} = \frac{1}{c}$. But $\sum_{x=1}^{\infty} \frac{1}{x} = \infty$, so there cannot be such a constant c .

3.2 Continuous Distributions

Commentary

This section ends with a brief discussion of probability distributions that are neither discrete nor continuous. Although such distributions have great theoretical interest and occasionally arise in practice, students can go a long way without actually concerning themselves about these distributions.

Solutions to Exercises

1. We compute $\Pr(X \leq 8/27)$ by integrating the p.d.f. from 0 to $8/27$.

$$\Pr\left(X \leq \frac{8}{27}\right) = \int_0^{8/27} \frac{2}{3}x^{-1/3}dx = x^{2/3}\Big|_0^{8/27} = \frac{4}{9}.$$

2. The p.d.f. has the appearance of Fig. S.3.1.

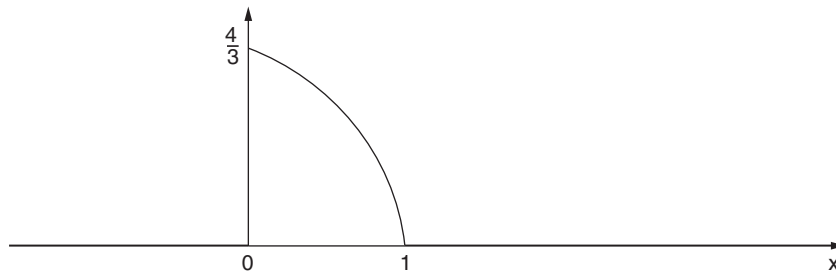


Figure S.3.1: Figure for Exercise 2 of Sec. 3.2.

(a) $\Pr\left(X < \frac{1}{2}\right) = \int_0^{1/2} 4(1-x^3)dx/3 = 0.6458.$